

Dingling Yao

Final-year ELLIS PhD Candidate in Computer Science, specializing in **interpretable representation learning for scientific discovery**. Experience with scalable Python/JAX/PyTorch workflows; published at NeurIPS, ICML, and ICLR.

Email: yaodingling806@gmail.com **Website:** ddcoan.github.io **GitHub:** github.com/DDCoan

LinkedIn: linkedin.com/in/dingling-yao **Google Scholar (h-index: 5, citations: 219):** scholar.google.com/dingling-yao

Education

ELLIS PhD in Computer Science

2022 – Present

Institute of Science and Technology Austria (ISTA) and Max Planck Institute for Intelligent Systems (MPI-IS) Austria & Germany

Topic: Interpretable Causal Representation Learning for Scientific Discovery.

PhD Advisor: Prof. Francesco Locatello **Co-advisor:** Prof. Georg Martius

Visiting Researcher, California Institute of Technology (Caltech)

Jun 2026 – Sep 2026

Anima AI + Science Lab

Pasadena, CA, USA

Host: Prof. Anima Anandkumar

Research on reliable scientific foundation models under interventions and distribution shifts, with applications to climate modeling.

M.Sc. in Machine Learning

2020 – 2022

University of Tübingen

Tübingen, Germany

Grade: *Sehr gut (very good)*

Relevant courses: Probabilistic Machine Learning, Deep Learning, Time Series and Dynamical Systems.

Industrial Research Experience

Experience spanning **fundamental research** (Amazon) and **applied research** (BCAI), with joint publications at top-tier venues.

Amazon Web Services

Oct 2022 – Mar 2023

Research Intern

Tübingen, Germany

Investigated **causal learning algorithms** for intrinsic robot exploration, combining representation learning and control.

Implemented scalable deep learning architectures in PyTorch for high-dimensional sensorimotor data, improving sample efficiency.

Bosch Center for Artificial Intelligence

May 2020 – Jun 2021

Part-time ML Researcher

Renningen, Germany

Developed **probabilistic models** for experimental design and adaptive data collection in industrial settings.

Awards & Fellowships

Outstanding academic contributions, as shown by publications at top-tier conferences and industrial fellowships.

Google PhD Fellowship

2025-2027

Machine Learning and ML Foundations

Amazon-MPI Fellowship

2022-2023

Amazon-MPI Science Hub

Top 3% paper

May 2024

ICLR 2024

Vienna, Austria

- First-author paper [8] selected as **Top 3%** at ICLR 2024.

Development Call Award

2026

EuroHPC

Italy

- European-wide HPC award

Leadership

NeurIPS Workshop Organization

2025

Lead Organizer

CauScien Workshop at NeurIPS 2025

Successfully organized the **first** CauScien Workshop at NeurIPS 2025 with more than 300 participants.

Speaker Series

2025

Organizer

ELLIS x UniReps

Co-organizing the ELLIS x UniReps Speaker Series on unifying representations with **over 100** participants.

Technical Skills

Climate & Scientific ML: Bulk ERA5/CMIP retrieval & preprocessing (CDS API), lightweight atmospheric modeling (SpeedyWeather.jl), scalable workflows with xarray/Zarr/WebDataset for multi-TB datasets

Languages & Frameworks: Python, PyTorch, JAX, scikit-learn, Julia, SQL

ML Expertise: Interpretable Representation Learning, Causal Inference, Generative Models, Reinforcement Learning (basic)

Cloud & Infrastructure: AWS, S3, HPC clusters

Invited Talks

Dedicated to impactful international research communication, evidenced by worldwide invited talks.

Causal Representation Learning for Scientific Discovery

Apr 2025

Institute of Computing Technology, Chinese Academy of Sciences

Remote

Multi-view Causal Representation Learning with Partial Observability

Mar 2024

SIAM Conference on Uncertainty Quantification (UQ24)

Trieste, Italy

Invited Talk

Feb 2024

Bellairs Workshop on Causality

Barbados

Oral Presentation

Dec 2023

Causal Representation Learning workshop at NeurIPS 2023

New Orleans, US

Academic Services

Reviewer

NeurIPS, ICML, ICLR, KDD AI for Science Track, TMLR, JCP, IEEE TNNLS

- Reviewed manuscripts for major machine learning, AI for science, and computational physics venues.

Publications and Preprints

My research focuses on developing scalable methods for interpretable representation learning from high-dimensional, real-world scientific datasets. By learning compact and semantically meaningful representations, my work enhances model interpretability and enables robust downstream scientific analyses across diverse domains, including climate science [7] and ecological studies [3].

Publications

- [1] **D. Yao**, A. Polesello, A. Pervez, C. Muller, and F. Locatello. “The Perception–Physics Paradox: Probing Scientific Alignment with TC-Bench”. In: *International Conference on Machine Learning (ICML 2026)*. 2026. URL: <https://github.com/CausalLearningAI/tc-bench>.
- [2] J. Chen, **D. Yao**, A. Pervez, D. Alistarh, and F. Locatello. “Scalable Mechanistic Neural Networks”. In: *International Conference on Learning Representations (ICLR 2025)*. 2025. URL: <https://arxiv.org/abs/2410.06074>.
- [3] **D. Yao**, D. Rancati, R. Cadei, M. Fumero, and F. Locatello. “Unifying Causal Representation Learning with the Invariance Principle”. In: *International Conference on Learning Representations (ICLR 2025)*. 2025. URL: <https://arxiv.org/abs/2409.02772>.
- [4] **D. Yao**, F. Tronarp, and N. Bosch. “Propagating Model Uncertainty through Filtering-based Probabilistic Numerical ODE Solvers”. In: *International Conference on Probabilistic Numerics 2025 (ProbNum 2025, oral)*. 2025. URL: <https://www.arxiv.org/abs/2503.04684>.
- [5] **D. Yao***, S. Huang*, R. Cadei, K. Zhang, and F. Locatello. “The Third Pillar of Causal Analysis? A Measurement Perspective on Causal Representations”. In: *Advances in Neural Information Processing Systems (NeurIPS 2025)*. 2025. URL: <https://arxiv.org/abs/2505.17708>.
- [6] D. Xu, **D. Yao**, S. Lachapelle, P. Taslakian, J. von Kügelgen, F. Locatello, and S. Magliacane. “A sparsity principle for partially observable causal representation learning”. In: *International Conference on Machine Learning (ICML 2024)*. 2024. URL: <https://openreview.net/pdf?id=MKG rRVODWR>.
- [7] **D. Yao**, C. Muller, and F. Locatello. “Marrying Causal Representation Learning with Dynamical Systems for Science”. In: *Advances in Neural Information Processing Systems (NeurIPS 2024)*. 2024. URL: <https://arxiv.org/abs/2405.13888>.

- [8] **D. Yao**, D. Xu, S. Lachapelle, S. Magliacane, P. Taslakian, G. Martius, J. von Kügelgen, and F. Locatello. “Multi-view causal representation learning with partial observability”. In: *International Conference on Learning Representations (ICLR 2024, spotlight, top 5%)*. 2024. URL: <https://openreview.net/forum?id=OGtnhKQJms>.
- [9] H. S. A. Yu, **D. Yao**, C. Zimmer, M. Toussaint, and D. Nguyen-Tuong. “Active learning in Gaussian process state space model”. In: *Machine Learning and Knowledge Discovery in Databases. Research Track: European Conference, ECML PKDD 2021, Bilbao, Spain, September 13–17, 2021, Proceedings, Part III* 21. Springer. 2021, pp. 346–361. URL: <https://arxiv.org/abs/2108.00819>.

*Equal contribution.